Week1 - Questions for Computer Vision

Concept Recall

1. What is the difference between a grayscale image and a color image in terms of matrix representation?
2. In a grayscale image, what do pixel values 0 and 255 represent?
3. How many channels does an RGB image have?
4. What is the coordinate system origin in computer vision images?
5. How do you access the pixel at position (x=10, y=5) in OpenCV?
6. Which color order does OpenCV use by default?
7. What does HSV stand for?
8. In HSV, what does the Hue component represent?
9. What does Saturation in HSV indicate?
10. What does Value in HSV indicate?
11. Define a kernel in image processing.
12. What is the purpose of convolution in image filtering?
13. What does a histogram represent in image processing?
14. What is thresholding used for?
15. Name two basic morphological operations.

Application & Explanation

1. Explain why HSV is often preferred over RGB for color detection tasks.
2. Give an example of a pixel value in RGB that represents pure red.
3. What happens when you apply a blur kernel to an image?
4. How does an edge detection kernel work?
5. Compare global thresholding and adaptive thresholding.
6. What information can be derived from an image histogram about brightness and contrast?
7. How does dilation differ from erosion in morphological operations?

8. Explain the difference between opening and closing operations.
9. Why are structuring elements important in morphological operations?
10. What is a convex hull in contour analysis?
11. How can bounding boxes be used in object localization?
12. Why is the origin (0,0) at the top-left corner in image representation?
13. How does Matplotlib's RGB order differ from OpenCV's BGR order?
14. What is the role of kernels in deep learning CNNs compared to classical image processing?
15. Explain how thresholding can be applied in medical imaging.

Analytical & Design-Oriented

1. Derive the mathematical expression for 2D convolution.
2. Discuss the limitations of global thresholding in images with uneven illumination and propose solutions.
3. How can histogram equalization improve image contrast?
4. Design a pipeline to detect ripe fruit using HSV color space.
5. Explain how contour approximation (Douglas-Peucker algorithm) can be used for shape classification.
6. How can morphological operations be combined to clean noisy scanned documents?
7. Propose a method to separate overlapping objects using thresholding, morphological operations, and contour analysis.
8. Compare the effectiveness of RGB vs HSV in detecting traffic lights under varying lighting conditions.
9. How does kernel size affect the output of convolution operations?
10. Explain how convolution is related to feature extraction in CNNs.
11. How can histograms be used to detect underexposed or overexposed images?
12. Discuss the trade-offs between using small vs large structuring elements in morphological operations.
13. How can contour features (area, perimeter, centroid) be used for object classification?
14. Propose a method to detect cracks in industrial materials using convolution and morphological operations.

15. Explain how combining HSV segmentation with contour analysis can improve object detection accuracy.